

# Does Structured Triage Context Help a Chest-Radiograph Pneumonia Model? A Time-Safe, Replicated, Externally Validated Multimodal Study with Cautions on Calibration and Missing Data

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**Abstract Background:** On arrival at the emergency department (ED), triage nurses record vital signs before any chest imaging is ordered. This has motivated multimodal models that combine the chest radiograph (CXR) with clinical context to detect pneumonia. It remains unclear whether such fusion actually helps once the features are restricted to what was available at the moment of prediction and the results are repeated across training runs, because earlier studies often report a single run and omit audits for leakage and missing data. **Methods:** We built a cohort that respects timing by linking MIMIC-CXR-JPG, MIMIC-IV and MIMIC-IV-ED, anchoring every feature to the moment the radiograph was acquired ( $t_0$ ) and admitting only measurements resolved at or before  $t_0$ . Splitting by patient and by time (test  $n=1,075$ ; prevalence 45.3%), we trained an image model, concatenation and attention fusion models, and clinical baselines over five random seeds, and added an early

laboratory panel as a third modality. We present the procedure as a reusable protocol, TEMPO (time-safe, equivalence-tested, missingness-audited, patient-level, out-of-sample). Alongside AUROC, AUPRC, expected calibration error (ECE) and Brier score, reported as mean $\pm$ SD over seeds with paired deltas and patient-level bootstrap intervals, we test equivalence against the margin with two one-sided tests (TOST) and report decision-curve, subgroup, interpretability (SHAP and Grad-CAM) and external analyses on NIH ChestX-ray14, together with explicit temporal, split and preprocessing audits. **Results:** A fine-tuned DenseNet-121 reaches AUROC  $0.737 \pm 0.003$ , well above the triage baselines (0.61). Adding triage vitals leaves discrimination unchanged ( $\Delta$ AUROC  $+0.004 \pm 0.009$ ; TOST confirms equivalence to the image model against a pre-specified  $\pm 0.05$  margin, with 80% power to detect a difference of 0.010), and a late-fusion design in the style of MedFuse behaves the same way. A calibration advantage that looked large in one run (ECE 0.040 vs 0.067) falls to  $\Delta$ ECE  $-0.013 \pm 0.016$  once averaged over seeds, so a single run overstated it roughly twofold. Early labs give a small but consistent gain ( $+0.009 \pm 0.005$ ) that an ablation traces entirely to which labs were ordered rather than to their values, and that is not leakage. On external data the image model transfers with little loss (AUROC  $0.722 \pm 0.005$  on 112,120 NIH radiographs). **Conclusion:** The radiograph carries the signal that separates pneumonia from its absence; the clinical context available at triage does not reliably add to it, and a naive fusion model can look helpful for reasons that will not hold up. We draw two cautions for clinical multimodal AI: a single training run can inflate an apparent calibration gain, and a model can lean on the pattern of which

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measurements are missing and pass it off as a benefit of the added modality.

**Keywords** Pneumonia detection · Chest radiograph · Multimodal fusion · Emergency department triage · Reasoning under uncertainty · Calibration · Informative missingness · Equivalence testing · Reproducibility · External validation

## 1 Introduction

Pneumonia is an acute lower respiratory tract infection and one of the leading reasons for emergency department (ED) presentation, hospital admission and death worldwide (Metlay et al, 2019). The frontal chest radiograph (CXR) is the recommended first-line investigation, yet its value in the ED is limited by how quickly a radiologist can read it during busy periods. An automated reader that returns well calibrated probabilities is therefore appealing for decision support, and deep convolutional networks now approach expert performance on many radiographic findings (Huang et al, 2017).

An appealing but underused source of complementary information is the ED triage record. On arrival a nurse records vital signs (temperature, heart rate, respiratory rate, peripheral oxygen saturation (SpO<sub>2</sub>), blood pressure and a pain score) together with an acuity level, all *before* imaging is ordered. Combining the radiograph with this readily available context is widely proposed (Acosta et al, 2022; Hayat et al, 2022; Soenksen et al, 2022), on the intuition that the two together should better reflect the clinical picture at the moment of decision. The intuition mirrors how clinicians work: a physician reads the image alongside the patient’s physiology rather than in isolation. Whether a model gains the same benefit is an empirical question, and it depends on how much information each modality carries that the other does not.

Three recurring problems in this literature motivate our study. The first is timing: clinical variables are often included without checking that they were actually available when the prediction is made, which risks leaking information generated after the decision point. The second is reliance on a single run: calibration and small differences in discrimination are frequently reported from one training seed, even though both are sensitive to training randomness (Guo et al, 2017; Nixon et al, 2019). The third is missing data: clinical inputs in the ED are missing often and not at random, so a model can learn from *whether* a measurement was taken rather than from its value, which can look like a genuine benefit of the extra modality.

We therefore ask a deliberately conditional question: *when, if ever, does the clinical context available at triage help a CXR pneumonia model?* We answer it under a verified timing constraint, with replication across seeds and explicit audits for missing data and leakage, following current reporting guidance for prediction models and medical imaging (Collins et al, 2024; Tejani et al, 2024). Rather than a new architecture, our contribution is a careful and reproducible evaluation, validated on external data, which we call **TEMPO** (Time-safe, Equivalence-tested, Missingness-audited, Patient-level, Out-of-sample; Algorithm 1). It adapts established reporting guidance to the particular hazards of clinical multimodal fusion, and it yields a clear answer together with two transferable cautions:

- A **time-safe, reproducible multimodal ED cohort** that links three MIMIC databases at the radiograph acquisition time  $t_0$ , with every feature checked to precede imaging (none of 314,145 laboratory values fall after  $t_0$ ).
- An **evaluation across five seeds** of the image model, triage fusion and laboratory fusion, compared with a paired bootstrap, which shows that the clinical context available at triage does not improve discrimination.
- A **rule of thumb for when fusion helps**: combining the radiograph with clinical data adds value when the target lies partly outside the image, as in prognosis or screening, but not when the image itself defines the label, as in radiographic diagnosis. This reconciles our negative result with the gains reported for prognostic tasks and indicates where fusion is worth the extra complexity.
- **Caution 1 (single-run calibration inflation)**: a calibration advantage that looks large from one training run shrinks by about half and becomes unstable once the run is repeated across seeds.
- **Caution 2 (missingness as signal)**: a small gain from laboratory data that an ablation reproduces entirely from the *missingness indicators*, with no contribution from the measured values, and that is not leakage.
- **External validation** of the image model on 112,120 NIH ChestX-ray14 radiographs, where it generalises across institutions with an AUROC drop of only 0.016, together with subgroup, decision-curve, interpretability and modern-backbone analyses.

## 2 Background and Related Work

### 2.1 Pneumonia and ED triage

Pneumonia manifests on a frontal CXR as regions of increased opacity (consolidation) where air-filled alveoli fill with exudate. Reported CXR sensitivity for pneumonia is moderate (0.43–0.74) with high specificity, reflecting both the difficulty of early detection and reliable exclusion in unambiguous cases (Metlay et al, 2019). Triage vital signs quantify physiological perturbation and severity; they are highly correlated with pneumonia but non-specific, since tachycardia, hypoxia and high acuity accompany many respiratory presentations. Crucially, triage measurements are recorded before imaging is ordered (a workflow guarantee, not a statistical regularity) so they carry no risk of temporal leakage.

### 2.2 The MIMIC database family

The Medical Information Mart for Intensive Care (MIMIC) programme provides the largest publicly available de-identified clinical dataset linked to real hospital records from the Beth Israel Deaconess Medical Center (BIDMC). This study uses three interoperable components. **MIMIC-CXR-JPG** (Johnson et al, 2019) contains 227,835 chest-radiograph studies from 64,588 patients (2011–2019) with CheXpert labels for 14 findings. **MIMIC-IV** (Johnson et al, 2023b) provides hospital-admission and laboratory records, enabling temporal consistency checks and the laboratory modality. **MIMIC-IV-ED** (Johnson et al, 2023a) provides ED-specific records for all BIDMC emergency visits, including the structured triage table used as the tabular input. All are distributed through PhysioNet under a Data Use Agreement; per-patient date shifts preserve temporal ordering and durations.

### 2.3 Automated labelling: CheXpert

CheXpert (Irvin et al, 2019) is an NLP pipeline that extracts binary labels for 14 thoracic conditions from free-text reports, assigning each a value of positive, negative, uncertain (−1), or not-mentioned. The uncertain label captures hedging language (“cannot exclude”, “possible pneumonia”). We adopt *u\_ignore* (exclude uncertain studies) as the primary policy, which preserves label quality at the cost of a smaller, higher-prevalence evaluation set.

### 2.4 Deep learning and multimodal fusion

DenseNet-121 (Huang et al, 2017) propagates features through dense connectivity,  $x_\ell = H_\ell([x_0, \dots, x_{\ell-1}])$ , and forms a strong CXR backbone under transfer learning. Intermediate (feature-level) fusion extracts modality-specific embeddings and combines them, by concatenation or by attention over modality tokens (Acosta et al, 2022).

Prior clinical fusion studies typically report positive gains, but on tasks that differ from ours in an important way. MedFuse fuses MIMIC clinical time-series with CXR for ICU mortality and phenotyping (Hayat et al, 2022), and the HAIM framework fuses imaging, tabular, text and time-series for inpatient outcomes (Soenksen et al, 2022); both target *outcomes* that are not directly visible in the radiograph. Recent emergency-department work is similar: Lee et al. (Lee et al, 2024) fuse CXR and EHR to screen for acute heart failure in the ED (AUROC 0.89), again a task whose label is not encoded in the image. Our result, that fusion is discrimination-neutral for radiographic *pneumonia*, is consistent with rather than contradictory to this body of work: pneumonia is defined *by* the radiograph, so the image already contains the label and there is little room for clinical context to add discriminative signal. This distinction, that fusion tends to help when the target lies partly outside the image but not when the image is essentially the target, is a useful lens on when multimodal CXR fusion should be expected to help.

Two further literatures frame our cautions. *Informative missingness*, the observation that missing values and their patterns correlate with the label, is well established and is often *exploited* as a predictive feature (Che et al, 2018); we demonstrate the flip side, that within a fusion model it can *masquerade* as a chemistry-driven benefit and therefore must be controlled for. *Calibration under distribution shift* is known to be fragile for single models and post-hoc scaling, with model averaging more robust (Ovadia et al, 2019); this is consistent both with our single-seed calibration caution and with the prevalence-driven degradation of ECE we observe on the external NIH set.

To our knowledge, no prior work simultaneously (i) targets ED-specific pneumonia, (ii) uses ED triage vitals under a verified time-safety constraint, (iii) evaluates with a patient-level temporal split *and* multi-seed replication, and (iv) audits missingness and leakage explicitly (Table 1). This study fills that gap.

**Table 1** Positioning against representative multimodal CXR + clinical-data studies. “Time-safe” denotes an explicitly verified constraint that all features precede the prediction time; “Missingness audit” denotes an explicit control isolating missingness from measured values; “Fusion gain” records whether the study reports multimodal fusion improving discrimination over its strongest single-modality baseline. ✓: reported; −: not reported.

| Study                        | Task               | Modalities          | Time-safe | Ext. val. | Multi-seed | Missing. audit | Fusion gain |
|------------------------------|--------------------|---------------------|-----------|-----------|------------|----------------|-------------|
| MedFuse (Hayat et al, 2022)  | ICU mortality      | CXR + time-series   | −         | −         | −          | partial        | reported    |
| HAIM (Soenksen et al, 2022)  | Inpatient outcomes | CXR + tab/text/TS   | −         | −         | −          | −              | reported    |
| Lee et al. (Lee et al, 2024) | ED heart-failure   | CXR + EHR           | −         | −         | −          | −              | reported    |
| <b>This work</b>             | ED pneumonia       | CXR + triage + labs | ✓         | ✓ (image) | ✓ (5)      | ✓ (flags)      | neutral     |

**Algorithm 1** TEMPO: time-safe, equivalence-tested, missingness-audited evaluation of clinical multimodal fusion.

**Require:** imaging + multi-source clinical records; per-instance prediction time  $t_0$ ; equivalence margin  $\Delta_0$ ; seed set  $S$

- 1: Anchor each instance at  $t_0$ ; admit a clinical feature only if its timestamp  $\leq t_0$ , and verify empirically that no feature post-dates  $t_0$  ▷ time-safety
- 2: Split at the *patient* level and temporally (order by earliest  $t_0$ ); fit all preprocessing on the training split only
- 3: **for** each seed  $s \in S$  **do**
- 4:   train the image-only and fusion models; store test predictions on identical rows
- 5: Aggregate per-seed metrics (mean  $\pm$  SD) and paired fusion–image deltas; quantify uncertainty by a patient-level paired bootstrap
- 6: **Equivalence:** TOST of the paired AUROC delta against  $\pm\Delta_0$ ; report the effect detectable at 80% power
- 7: **Missingness audit:** refit fusion with missingness *indicators only* (no values); attribute any gain to structure vs. measured values
- 8: Report calibration with multi-seed spread and bootstrap CIs; validate the transferable branch on an external dataset

**Ensure:** discrimination/calibration verdict with an equivalence decision and a missingness attribution

### 3 Materials and Methods

The study instantiates the TEMPO protocol (Algorithm 1); the following subsections detail each step, and Figure 1 shows the data flow it produces.

#### 3.1 Dataset description

The task is binary radiographic-pneumonia classification of a frontal ED CXR. Each of the three MIMIC components plays a distinct role. MIMIC-CXR-JPG supplies the *images* (JPEG radiographs) and the *labels* (CheXpert pneumonia), together with DICOM metadata (view position; StudyDate/StudyTime, from which the temporal anchor  $t_0$  is derived). MIMIC-IV-ED supplies the *primary tabular modality*: the triage table (vitals, pain, acuity) recorded at ED intake, linked to each CXR through the ED stay. MIMIC-IV supplies the *admission records* used to verify temporal

consistency and the *laboratory events* used for the time-safe laboratory modality. The image branch consumes MIMIC-CXR-JPG; the tabular branch consumes MIMIC-IV-ED (triage) and, in the laboratory variant, MIMIC-IV (labs); the CheXpert labels supervise all models.

#### 3.2 Cohort construction pipeline

A scripted pipeline links the sources through temporally constrained relational joins. Each CXR study is linked to a hospital admission (by `subject_id` and admission window) and to an ED stay (by `stay_id` with  $t_0 \in [\text{intime}, \text{outtime}]$ ); one-to-one study-to-stay integrity is enforced by deduplication. The triage table is merged on `stay_id`; studies are restricted to frontal (PA/AP) projections; CheXpert pneumonia labels are merged and the *u.ignore* policy applied. The temporal anchor is  $t_0 = \text{StudyDate} + \text{StudyTime}$  from DICOM metadata. The resulting cohort comprises 9,154 ED-anchored studies (Figure 1); the patient-level temporal split (below) assigns train 7,144 / validation 930 / test 1,080 studies. Models train and evaluate only on radiographs present on disk in our data snapshot, yielding a held-out test set of  $n=1,075$  (487 positive / 588 negative; prevalence 45.3%). The entire pipeline is reproducible from a single entry point, emitting versioned manifests at each stage. Table 2 summarises the cohort: the split sizes, test-set demographics (median age 65; 55% female; predominantly White with substantial Black and Hispanic representation), the near-even PA/AP view split, and the triage vital-sign distributions with their missingness rates (3–8% per vital), which motivate the per-vital missingness indicators.

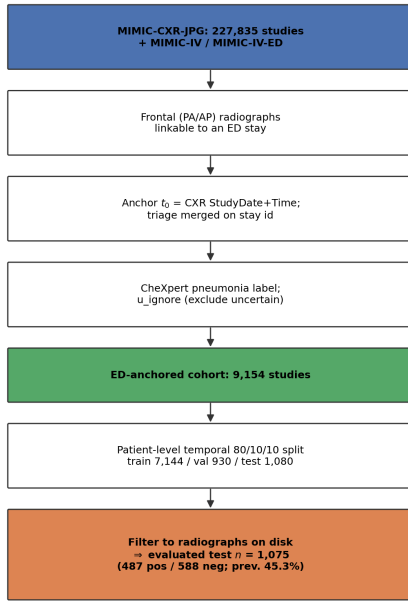
#### 3.3 Time-safety and the patient-level temporal split

Every clinical feature is either a triage measurement (recorded at intake, before imaging by protocol) or a laboratory value with charttime in  $(t_0 - 24\text{h}, t_0]$ . Time-safety was verified empirically on the extracted

**Table 2** Cohort characteristics. Splits are patient-level and temporally ordered; demographic and vital-sign statistics are for the test split.

| Characteristic                                 | Value                 |
|------------------------------------------------|-----------------------|
| <i>Splits: studies / patients / prevalence</i> |                       |
| Train                                          | 7,144 / 6,045 / 46.9% |
| Validation                                     | 930 / 787 / 48.8%     |
| Test                                           | 1,080 / 890 / 45.4%   |
| <i>Test-set demographics</i>                   |                       |
| Age, mean (median)                             | 64 (65)               |
| Female                                         | 55.0%                 |
| White / Black / Hispanic                       | 678 / 258 / 71        |
| Asian / Other                                  | 36 / 37               |
| View: PA / AP                                  | 56.8% / 43.2%         |

| Triage vital (test set) | mean±SD (% missing) |
|-------------------------|---------------------|
| Temperature (°F)        | 98.5±1.4 (5.9%)     |
| Heart rate (bpm)        | 87.7±19.6 (3.3%)    |
| Respiratory rate (/min) | 18.6±3.5 (4.5%)     |
| SpO <sub>2</sub> (%)    | 97.4±3.6 (4.8%)     |
| Systolic BP (mmHg)      | 135±24 (4.0%)       |
| Diastolic BP (mmHg)     | 73±16 (4.3%)        |
| Pain (0–10)             | 3.8±4.3 (8.2%)      |
| Acuity (ESI 1–5)        | 2.3±0.7 (1.2%)      |

**Fig. 1** Cohort-construction flow from the full MIMIC-CXR-JPG collection to the ED-anchored, time-safe evaluation set. Coloured boxes mark the raw sources (blue), the linked ED-anchored cohort (green) and the held-out test set actually evaluated on disk (orange).

laboratory events: *zero* of 314,145 measurements have a charttime after  $t_0$  (median lead time 1 h). Patients are ranked by their earliest  $t_0$  and partitioned 80/10/10 into train/validation/test, so no patient appears in more than one split (verified: 0 cross-split subjects, 0 duplicate studies). This ordinal-time construction approximates prospective deployment. The tabular preprocessor (median imputation, standardisation, one-hot encoding) is fit on the training split only.

### 3.4 Feature engineering

**Triage features** are temperature, heart rate, respiratory rate, SpO<sub>2</sub>, systolic and diastolic blood pressure, pain score and ESI acuity, plus the radiograph view and demographics. Physiologically implausible values are clipped to clinical ranges. For each vital a binary *missingness indicator* is added and the value median-imputed; because ED missingness is not missing-at-random, the indicator itself carries information. **Laboratory features** comprise 26 concepts (complete blood count, metabolic panel, liver panel, blood gas, lactate, and the inflammatory markers CRP and procalcitonin). For each study and concept we take the last value resolved at or before  $t_0$  within the 24 h window, and add a missingness flag recomputed after the cohort join.

### 3.5 Models

Figure 2 shows the multimodal architecture. The **image model** is DenseNet-121, initialised from ImageNet, multilabel-pretrained on 182,637 non-ED MIMIC-CXR studies (14 CheXpert findings, masked binary cross-entropy over observed labels), then fine-tuned for binary pneumonia at  $224 \times 224$  with class-balanced loss and differential learning rates ( $5 \times 10^{-5}$  head,  $10^{-5}$  backbone). Multilabel pretraining minimises a masked binary cross-entropy that ignores not-mentioned CheXpert labels,

$$\mathcal{L}_{\text{mBCE}} = - \sum_i \frac{1}{m_i} \sum_i m_i [y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i)], \quad (1)$$

where  $m_i \in \{0, 1\}$  masks unobserved conditions. The triage vector is encoded by a two-layer TabularMLP (batch-normalised, dropout 0.2) to a 128-d embedding

$e_t$ . The **concatenation** model concatenates the 1024-d image embedding  $e_v$  with  $e_t$ ,  $[e_v; e_t] \in \mathbb{R}^{1152}$ , and applies a fully connected head; the **attention** model projects each embedding to a  $d$ -dimensional token, prepends a learnable CLS token, and applies a two-layer Transformer whose scaled dot-product attention

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \quad (2)$$

mixes the  $[\text{CLS}, e_v, e_t]$  tokens; the CLS output feeds the classifier (Acosta et al, 2022). The **+labs** model appends the 26 lab values and their flags to the tabular vector. **Clinical baselines** are logistic regression and XGBoost (Chen and Guestrin, 2016) on triage features. A **late-fusion baseline** fuses at the decision level rather than the feature level: the independently trained image and triage-logistic models each emit a probability and a logistic meta-learner combines them, fit on the validation split, a static analogue of the MedFuse late-fusion design (Hayat et al, 2022). A **modern-backbone baseline** uses a CheXpert-trained DenseNet-121 (Cohen et al, 2022) (not trained on MIMIC), zero-shot and as a frozen-feature linear probe. All deep models share the multilabel-pretrained backbone so the multi-seed analysis isolates fine-tuning stochasticity.

### 3.6 Training and evaluation

Each deep model is trained across five random seeds (42, 123, 456, 789, 1000) with early stopping on validation AUPRC. Primary metrics are AUROC, AUPRC, ECE and Brier score, reported as mean $\pm$ SD over seeds with paired per-seed deltas against the image-only model on identical test rows; within-checkpoint uncertainty uses a 2,000-replicate patient-level paired bootstrap (patients, not studies, are resampled). A symmetric equivalence margin of  $\pm 0.05$  AUROC was pre-specified (Piaggio et al, 2012); equivalence of a fusion model to the image-only model is tested formally by two one-sided tests (TOST) against  $\pm 0.05$  on the patient-level paired-bootstrap distribution of the AUROC delta, and we report the minimal effect detectable at 80% power. Calibration is assessed with the ECE

$$\text{ECE} = \sum_{b=1}^B \frac{n_b}{n} |\bar{p}_b - \bar{y}_b|, \quad (3)$$

(with  $B=10$  uniform bins;  $\bar{p}_b, \bar{y}_b$  the mean predicted probability and observed frequency in bin  $b$ ), the maximum calibration error, Brier score and reliability diagrams; ECE bin-count/scheme sensitivity is examined.

Clinical utility uses decision-curve analysis with net benefit

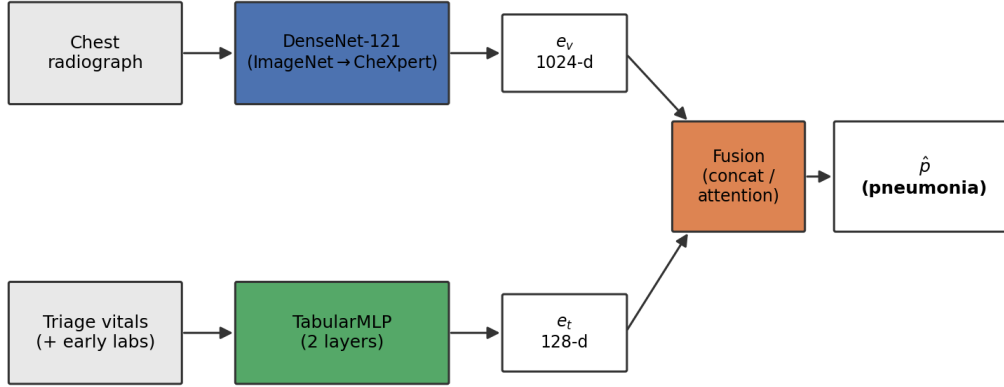
$$\text{NB}(p_t) = \frac{\text{TP}}{n} - \frac{\text{FP}}{n} \cdot \frac{p_t}{1 - p_t}. \quad (4)$$

Interpretability uses exact TreeSHAP (Chen and Guestrin, 2016) for the XGBoost baseline and Grad-CAM for the image branch. We additionally compute subgroup metrics (sex, age, race, acuity, view), operating points at fixed sensitivity, a laboratory *missingness-only* ablation, and external validation on all 112,120 NIH ChestX-ray14 radiographs (Wang et al, 2017) with preprocessing matched to the internal transform.

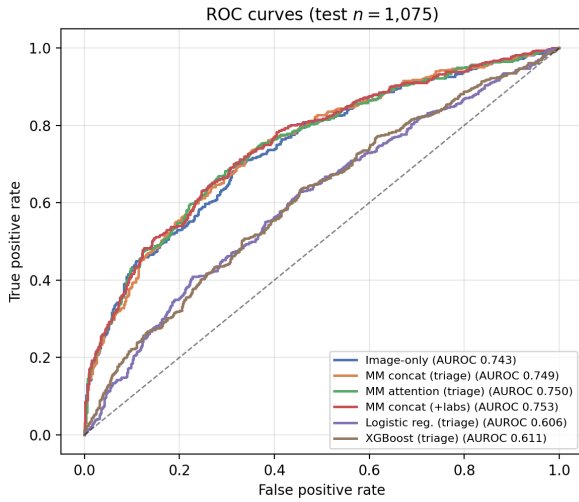
## 4 Results

### 4.1 Discrimination: imaging dominates, fusion is neutral

Table 3 and Figs. 3–5 summarise the primary results. Both CXR models beat the triage baselines by more than 12 AUROC points, which confirms that the radiograph carries the signal that discriminates. Adding triage vitals does not change discrimination: across seeds the paired concat–image  $\Delta\text{AUROC}$  is  $+0.004 \pm 0.009$  (range  $[-0.007, +0.014]$ ). On a paired bootstrap over patients of the seed-averaged models the delta is  $+0.006$ , with a 90% confidence interval of  $[+0.000, +0.012]$  that lies entirely inside the pre-specified  $\pm 0.05$  margin; both one-sided tests reject at  $p < 0.001$ , so fusion is formally equivalent to imaging in discrimination. Because the two models share a backbone and their predictions are strongly correlated, the paired comparison has good power: the smallest AUROC difference we could detect at 80% power is 0.010, five times below the margin, so the null is not an artefact of low power. The result does not depend on our choice of feature-level fusion either. A late-fusion baseline in the style of MedFuse, which combines the separately trained image and triage models with a logistic meta-learner fitted on the validation split, is also neutral ( $\Delta\text{AUROC} +0.000 \pm 0.001$  across seeds; TOST again confirms equivalence), even though the meta-learner gives the triage stream a real weight. No fusion we tried, whether early or late, recovers a gain in discrimination. Attention fusion matches the image AUROC but is badly and unevenly calibrated (ECE  $0.076 \pm 0.042$ ; Fig. 6), a reminder that AUROC alone is not enough for choosing a model. The precision-recall curves (Fig. 4) follow the same ordering at the 45.3% prevalence. The result also shows in the representations: in a t-SNE of the learned embeddings (Fig. 7), the triage embedding is not aligned with the label



**Fig. 2** Multimodal architecture. A DenseNet-121 image encoder (ImageNet then CheXpert multilabel pretraining) produces a 1024-d embedding  $e_v$ ; a two-layer TabularMLP encodes triage vitals (optionally with early labs) to a 128-d embedding  $e_t$ . The streams are combined by concatenation or by CLS-token attention and passed to a classifier head that outputs the pneumonia probability.

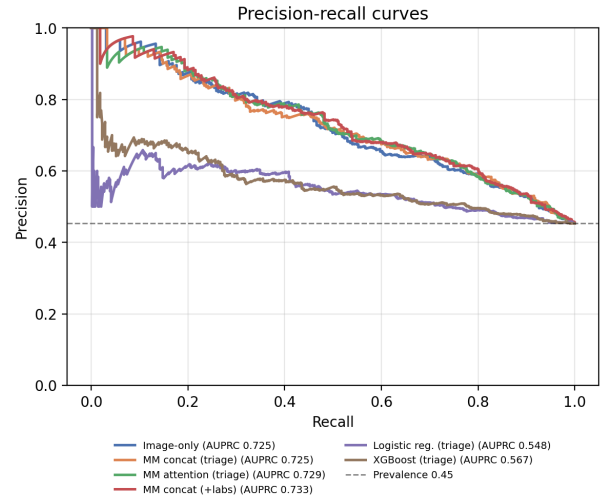


**Fig. 3** ROC curves on the held-out test set (5-seed mean predictions for the deep models; AUROC of the ensemble curve shown, slightly above the per-seed means in Table 3).

and the fused representation is essentially the image embedding.

#### 4.2 Feature ablation

Retraining the concat model with restricted triage feature subsets (Table 4, representative seed 42) leaves AUROC essentially unchanged (0.736–0.752, a 0.016 range), confirming that the CXR backbone, not any



**Fig. 4** Precision-recall curves at test prevalence 45.3%.

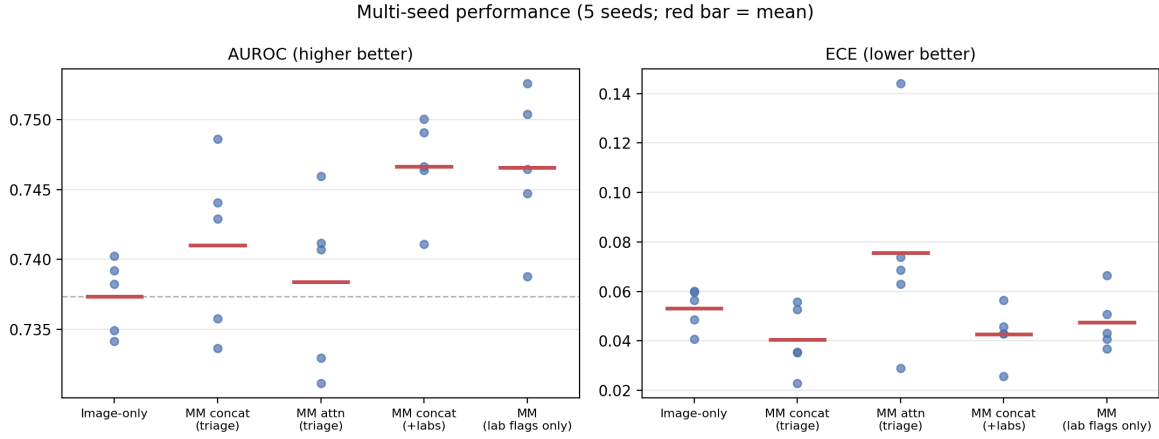
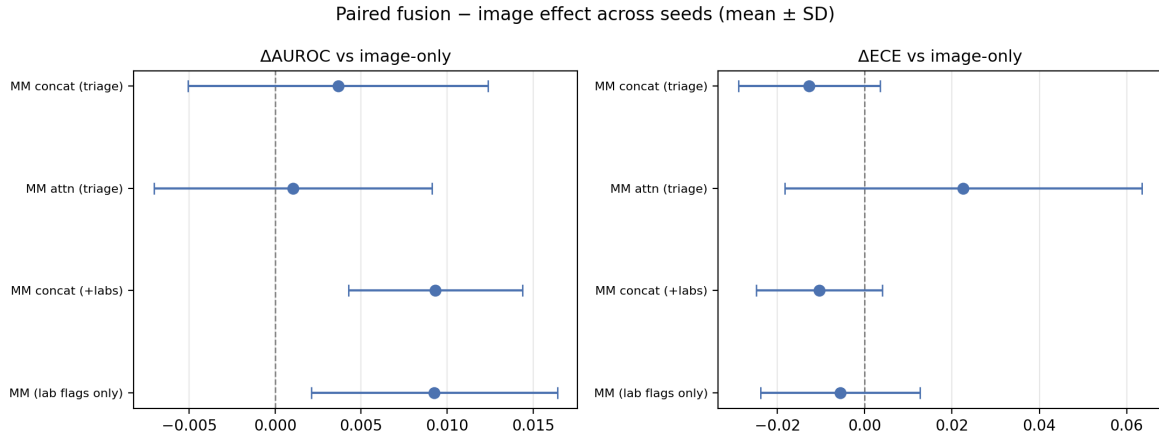
particular triage feature, drives discrimination. Removing the missingness indicators does not reduce performance, consistent with their marginal, largely missingness-pattern contribution (Section 4.4).

#### 4.3 Caution 1: single-seed evaluation inflates the calibration effect

A single checkpoint suggested the concat model was markedly better calibrated (ECE 0.040 vs 0.067, a 40% reduction). Across seeds, however, the image-only

**Table 3** Multi-seed test-set performance (mean $\pm$ SD over 5 seeds;  $n=1,075$ ; prevalence 45.3%). Clinical baselines are single-fit.

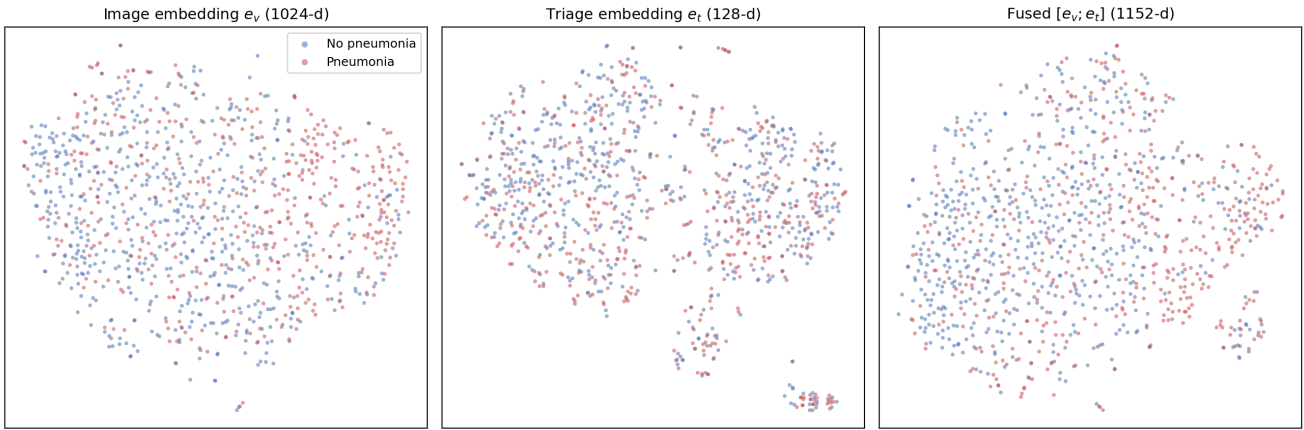
| Model                          | AUROC             | AUPRC             | ECE               | Brier             |
|--------------------------------|-------------------|-------------------|-------------------|-------------------|
| Image-only DenseNet-121        | 0.737 $\pm$ 0.003 | 0.719 $\pm$ 0.004 | 0.053 $\pm$ 0.008 | 0.207 $\pm$ 0.001 |
| Multimodal (concat, triage)    | 0.741 $\pm$ 0.006 | 0.715 $\pm$ 0.005 | 0.040 $\pm$ 0.014 | 0.205 $\pm$ 0.002 |
| Multimodal (attention, triage) | 0.738 $\pm$ 0.006 | 0.713 $\pm$ 0.012 | 0.076 $\pm$ 0.042 | 0.212 $\pm$ 0.010 |
| Multimodal (concat, +labs)     | 0.747 $\pm$ 0.004 | 0.724 $\pm$ 0.004 | 0.043 $\pm$ 0.011 | 0.203 $\pm$ 0.002 |
| Logistic regression (triage)   | 0.606             | 0.548             | 0.037             | 0.242             |
| XGBoost (triage)               | 0.611             | 0.567             | 0.046             | 0.241             |

**Fig. 5** Per-model AUROC and ECE across five seeds (points, seeds; red bar, mean). Discrimination is stable; the image-only ECE alone varies substantially by seed.**Fig. 6** Paired fusion—image effects across seeds (mean $\pm$ SD). Every effect straddles zero; the labs and lab-flags-only deltas are identical.**Table 4** Triage feature-group ablation for the concat model (seed 42). AUROC is stable across subsets, confirming the CXR backbone dominates.

| Triage features                           | AUROC | AUPRC |
|-------------------------------------------|-------|-------|
| All (vitals, acuity, demographics, flags) | 0.736 | 0.714 |
| Vitals only                               | 0.740 | 0.714 |
| Vitals + acuity                           | 0.744 | 0.723 |
| No missingness flags                      | 0.752 | 0.734 |

ECE alone ranges 0.041–0.060, and the paired advantage shrinks to  $\Delta\text{ECE} -0.013 \pm 0.016$  (4/5 seeds favour concat; the range crosses zero). The single-seed estimate thus overstated the benefit roughly twofold. Within-checkpoint uncertainty tells the same story: patient-level bootstrap 95% CIs on the seed-averaged ECE overlap almost entirely (image 0.042, CI [0.033, 0.077]; concat 0.037, CI [0.028, 0.068]), so the calibration gap is not resolvable even before seed variance is added. The apparent gap is also binning-



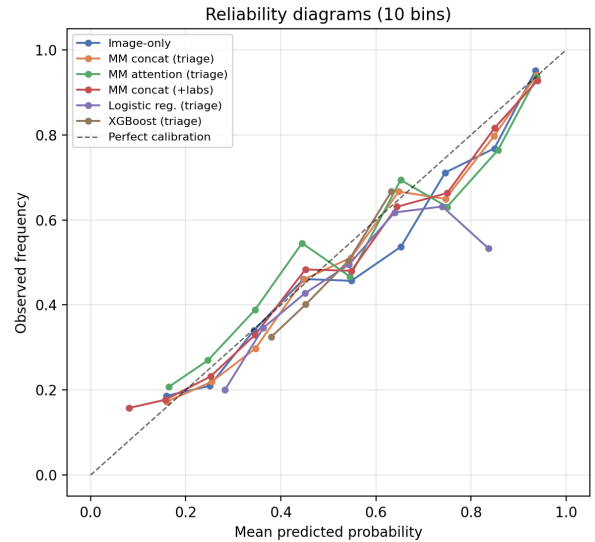


**Fig. 7** t-SNE of the learned representations on the test set ( $n=1,075$ ), coloured by pneumonia label. The image embedding  $e_v$  (left) carries a moderate label gradient, consistent with its AUROC. The triage embedding  $e_t$  (centre) forms its own structure driven by vital-sign patterns, but that structure is not aligned with the label. The fused vector  $[e_v; e_t]$  (right) is essentially the image embedding: appending the triage stream does not reorganise the representation into better class-separated geometry, the representational counterpart of the fusion-neutral discrimination result.

dependent (Fig. 9): under coarser or quantile bins it shrinks further (Nixon et al, 2019). Brier scores are essentially unchanged, so any effect is distributional alignment rather than refinement. Reliability diagrams (Fig. 8) show all deep models tracking the diagonal closely, with attention fusion the clear outlier. Finally, a simple post-hoc correction removes what remains of the gap: temperature scaling of the image model (optimal  $T = 1.37$ , fit on the validation logits) lowers its test ECE from 0.060 to 0.042, matching the concat model’s 0.040. The modest calibration difference therefore does not motivate multimodal fusion on calibration grounds, since a single-parameter recalibration of the image model alone recovers it (Ovadia et al, 2019).

#### 4.4 Caution 2: the “labs” benefit is missing-data structure

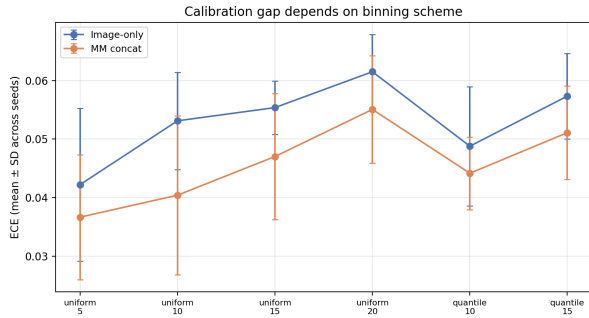
Appending early labs gives the only seed-consistent AUROC gain (labs+image  $+0.009 \pm 0.005$ , 5/5 seeds, range excluding zero). Three analyses show this is neither chemistry nor leakage. *Ablation*: a model using laboratory *missingness flags only* (no values) reproduces the gain exactly (flags+image  $+0.009$ ; Fig. 6); the lab values add nothing further to discrimination. *Subcohort*: on the 269 test studies that actually have labs, discrimination is tied (labs 0.778 vs image 0.776). *Leakage audit*: 0 of 314,145 lab measurements post- $t_0$ ; 0 cross-split subjects; preprocessing fit on train only; the “any lab drawn” indicator has near-null univariate AUROC (0.464). The gain therefore reflects the weak complementary signal of *whether* bloodwork was ordered, is small, and lies within the patient-level bootstrap CI.



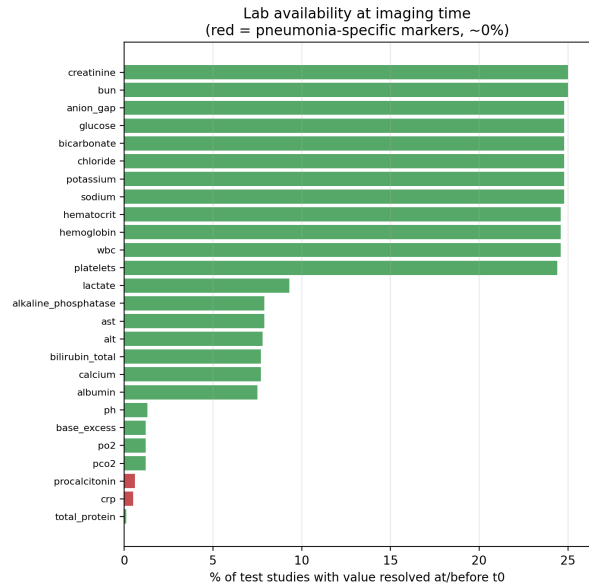
**Fig. 8** Reliability diagrams (10 bins). The deep models track the diagonal; attention fusion systematically under-predicts.

#### 4.5 Clinical availability of laboratory results

Under time-safe matching, only about a quarter of ED-anchored CXR studies have any resolved laboratory value at or before imaging (25.4% of the full cohort; 269 of the 1,075 test studies, 25.0%; Fig. 10). Complete-blood-count and metabolic panels reach  $\sim 25\%$ , but the most pneumonia-specific markers (CRP and procalcitonin) are present in only 0.5% and 0.6% of studies. At the moment of imaging the informative labs are essentially unavailable, which explains both the small labs effect and its missingness origin.



**Fig. 9** ECE under different binning schemes; the gap between the image and concat depends on the binning choice.



**Fig. 10** Fraction of test studies with each laboratory value resolved at or before  $t_0$ ; pneumonia-specific markers (red) are essentially unavailable at imaging time.

#### 4.6 Operating points and clinical utility

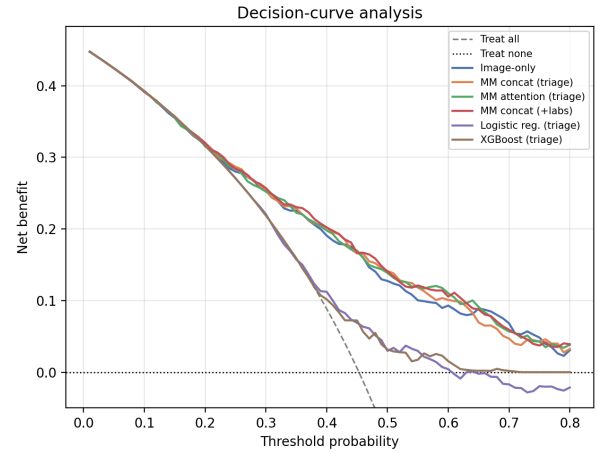
ED triage for pneumonia prioritises sensitivity. At a fixed sensitivity of 0.90 the specificity is 0.31 (image) versus 0.32 (concat); at 0.95, 0.17 versus 0.18 (Table 5), clinically equivalent operating behaviour. Decision-curve analysis (Fig. 11) shows both CXR models providing substantially higher net benefit than the triage-only baselines and the treat-all/treat-none references across the clinically relevant threshold range, with the image and multimodal curves essentially overlapping.

#### 4.7 Subgroup performance

Table 6 and Fig. 12 report subgroup AUROC. Performance is markedly lower for female than male patients

**Table 5** Operating points at fixed sensitivity (mean over 5 seeds).

| Model  | Sens. | Spec. | PPV   | NPV   |
|--------|-------|-------|-------|-------|
| Image  | 0.90  | 0.310 | 0.520 | 0.791 |
| Concat | 0.90  | 0.322 | 0.524 | 0.797 |
| Image  | 0.95  | 0.173 | 0.488 | 0.808 |
| Concat | 0.95  | 0.179 | 0.490 | 0.814 |

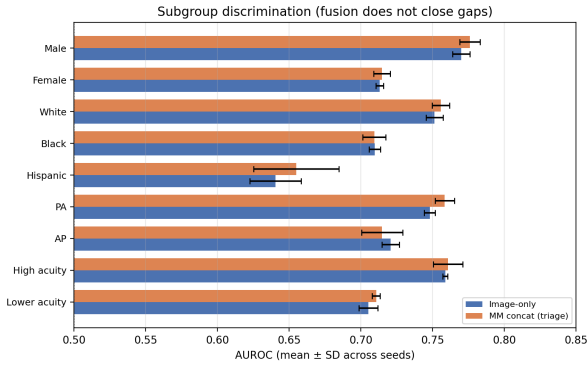


**Fig. 11** Decision-curve analysis: net benefit versus threshold probability. Imaging models dominate the triage baselines and reference strategies.

**Table 6** Subgroup AUROC (mean over 5 seeds) for image-only and concat fusion. Small strata (e.g. Hispanic, Asian) have wide CIs.

| Subgroup           | Image | Concat |
|--------------------|-------|--------|
| Female             | 0.713 | 0.715  |
| Male               | 0.770 | 0.776  |
| White              | 0.751 | 0.756  |
| Black              | 0.710 | 0.709  |
| Hispanic           | 0.640 | 0.655  |
| PA view            | 0.748 | 0.758  |
| AP view            | 0.721 | 0.715  |
| High acuity (1–2)  | 0.759 | 0.761  |
| Lower acuity (3–5) | 0.705 | 0.711  |

(0.71 vs 0.77) and for Hispanic patients (0.64) relative to White (0.75); PA views outperform AP (0.75 vs 0.72). Fusion does not consistently close these gaps and, in the smallest strata, can slightly worsen calibration. These disparities echo documented underdiagnosis biases in CXR models (Seyyed-Kalantari et al, 2021; Larrazabal et al, 2020) and are reported here as a pre-requisite fairness audit rather than a solved problem; small minority strata carry wide confidence intervals. These subgroup contrasts are exploratory and not corrected for multiple comparisons, so the individual gaps should be read as hypothesis-generating rather than as confirmatory effects.



**Fig. 12** Subgroup AUROC (mean $\pm$ SD across seeds); fusion does not close sex, race or view gaps.

**Table 7** Mean |SHAP| for the triage XGBoost model (test set).

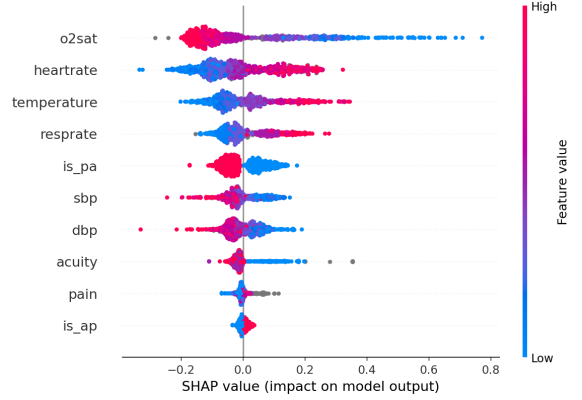
| Feature           | Mean  SHAP |
|-------------------|------------|
| SpO <sub>2</sub>  | 0.148      |
| Heart rate        | 0.099      |
| Temperature       | 0.078      |
| Respiratory rate  | 0.062      |
| PA-view indicator | 0.053      |
| Systolic BP       | 0.043      |
| Diastolic BP      | 0.042      |
| Acuity (ESI)      | 0.031      |
| Pain score        | 0.012      |

#### 4.8 Interpretability

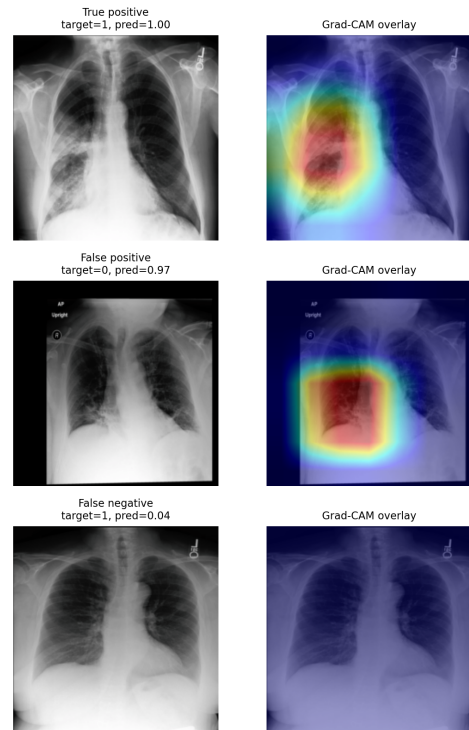
For the triage-only XGBoost model, exact TreeSHAP attributes the largest importance to SpO<sub>2</sub> (mean |SHAP| = 0.148), followed by heart rate (0.099), temperature (0.078) and respiratory rate (0.062) (Table 7; Fig. 13), consistent with the pathophysiology of pneumonia, in which alveolar consolidation impairs gas exchange. Grad-CAM on the image branch (Fig. 14) localises activation over regions of consolidation in confident true positives, whereas a false positive attends to overlapping opacity and a false negative shows diffuse, non-localised activation.

#### 4.9 Modern backbone and external validation

A CheXpert-trained DenseNet-121 (Cohen et al, 2022), used zero-shot (AUROC 0.667) or as a frozen-feature linear probe (0.687), does not outperform our task-tuned model (0.737), indicating the conclusions are not an artefact of an outdated backbone. To test whether fusion neutrality is instead specific to the CNN architecture or to feature freezing, we also fine-tuned a Vision Transformer (ViT-Small/16, ImageNet-pretrained) end-to-end on the identical cohort and split (seed 42):

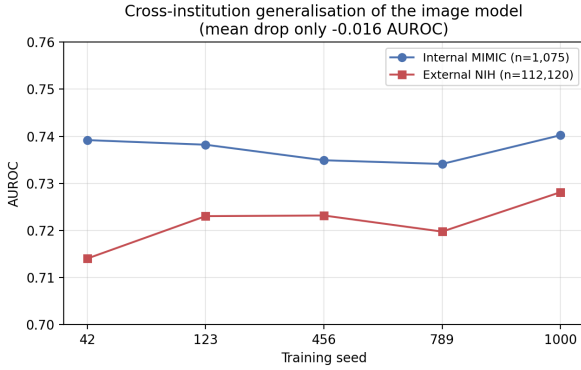


**Fig. 13** SHAP summary (beeswarm) for the triage XGBoost model; low SpO<sub>2</sub> pushes predictions toward pneumonia.



**Fig. 14** Grad-CAM on the image model for representative test cases: a confident true positive, a false positive, and a false negative.

image-only reached AUROC 0.712 and triage-concat fusion 0.696, a change of  $-0.015$ . A contemporary transformer backbone, fully fine-tuned, therefore also extracts no discriminative synergy from triage vitals over imaging alone, so the neutrality holds across both backbone family and training regime. On external validation against all 112,120 NIH ChestX-ray14 radiographs (Wang et al, 2017) (a different institution, scanner population and label pipeline; prevalence 1.28%), discrimination transfers with minimal loss: AUROC  $0.722 \pm 0.005$  across five seeds (ensemble 0.726), only 0.016 be-



**Fig. 15** Cross-institution generalisation: per-seed AUROC on internal MIMIC ( $n=1,075$ ) versus external NIH ChestX-ray14 ( $n=112,120$ ); mean drop  $-0.016$ .

**Table 8** Label-policy sensitivity (seed 42): AUROC/AUPRC for image and concat under three CheXpert uncertainty policies. Test size and prevalence differ by policy.

| Model  | Policy          | Test $n$ | AUROC | AUPRC |
|--------|-----------------|----------|-------|-------|
| Image  | <i>u_ignore</i> | 1,075    | 0.739 | 0.724 |
|        | <i>u_zero</i>   | 1,944    | 0.597 | 0.342 |
|        | <i>u_one</i>    | 1,944    | 0.761 | 0.873 |
| Concat | <i>u_ignore</i> | 1,075    | 0.736 | 0.714 |
|        | <i>u_zero</i>   | 1,944    | 0.595 | 0.357 |
|        | <i>u_one</i>    | 1,944    | 0.765 | 0.876 |

low the internal 0.737 (Fig. 15). As expected under the large prevalence shift ( $45\% \rightarrow 1.3\%$ ), prevalence-dependent metrics degrade (AUPRC 0.030, ECE 0.44) because probabilities are calibrated to the internal base rate; AUROC, which is prevalence-independent, is the appropriate cross-dataset metric, and it holds despite domain shift and NLP-derived external labels.

#### 4.10 Label-policy sensitivity

CheXpert uncertain labels can be excluded (*u\_ignore*, our primary policy), mapped to negative (*u\_zero*) or positive (*u\_one*); each changes the cohort size and prevalence (Table 8). The *u\_zero* policy sharply lowers AUROC and AUPRC because treating uncertain studies as negative injects label noise and false negatives, whereas *u\_one* inflates prevalence to 69.8% and thus raises both metrics; *u\_ignore* is intermediate and the most defensible. Critically, the image and concat models remain essentially equivalent under every policy, confirming that the discrimination-neutrality of fusion is not an artefact of the uncertainty-handling choice.

## 5 Discussion

The chest radiograph carries the discriminative signal for radiographic pneumonia in this ED cohort. Immediately available structured clinical context (triage vitals, and even a broad early-laboratory panel) does not add reliable discrimination, and the image model generalises externally with minimal loss. The asymmetry is intuitive: the radiograph encodes the outcome (radiographic consolidation) almost directly, whereas vital signs and labs encode severity and physiological perturbation that are correlated with, but not specific to, pneumonia.

Two findings carry beyond this particular task. First, a calibration advantage that looks large in a single run (a 40% reduction in ECE) is roughly halved and becomes unstable once we average over five seeds; calibration claims from one run should be read with caution, and replication across seeds ought to be routine. Second, when clinical inputs are missing often and not at random, a fusion model can exploit the pattern of missingness and show a benefit that reflects neither the measured values (the subcohort with labs present shows no effect of the values) nor leakage (the temporal, split and preprocessing audits are clean). The ablation that keeps only the missingness indicators makes this mechanism explicit, and to our knowledge it is rarely reported in clinical multimodal studies. Both cautions are easy and cheap to apply.

*A cognitive-computation view.* Fusion is, in effect, a machine version of clinical reasoning: it brings together different kinds of evidence, the image, the physiological signals, and even what was and was not measured, into one judgement under uncertainty. Seen this way, our two cautions are about how the machine reasons rather than about bookkeeping. The first concerns how far its stated confidence can be trusted: a calibration figure from one training run is a shaky guide, so its uncertainty has to be measured across repeated runs before it can inform a decision. The second concerns reasoning from absence: an ED record is missing for reasons that are not random, and a model readily treats the presence of a measurement as evidence, much as a clinician reads “a test was ordered” as informative, but it confuses that ordering signal with the physiology it is meant to capture. Knowing when combined evidence truly adds information, and when an apparent gain instead reflects unstable confidence or the structure of missingness, matters for cognitively inspired decision support well beyond this radiographic task.

*When does fusion help?* Our result sharpens a simple rule. Fusion of the radiograph with clinical data has

worked best for prognostic and screening targets whose ground truth is not the image itself, such as ICU mortality (Hayat et al, 2022), inpatient outcomes (Soenksen et al, 2022) and screening for acute heart failure in the ED (Lee et al, 2024), where the clinical record supplies information the image cannot. Radiographic pneumonia is the opposite case: the label is defined by the image, so triage vitals and early labs have little to add, and our result is what the rule predicts. The point is not that fusion fails, but that its value depends on the target lying partly outside the image, which is a useful guide to where fusion is worth the extra complexity in the ED.

*Label provenance and the ceiling on discrimination.*

Two related objections deserve a direct answer. First, the CheXpert labels come from the radiology report, so the image is by construction close to sufficient for the label. We see this not as a confound but as the mechanism behind our thesis: when the reference standard is read off the image, clinical context outside the image has little room to add, and a neutral result is what a well-posed task of this kind should give. If anything it makes the conclusion conservative, since a label taken from the report favours the image as much as possible. Second, the absolute AUROC (about 0.74) is moderate, which is expected: pneumonia is one of the CheXpert findings with the lowest reader agreement, below 40% expert agreement in the CheXpert study (Irvin et al, 2019), so part of the ceiling is genuine label noise rather than limited model capacity. That noise affects every model equally and cancels in the paired comparison, so it does not weaken the equivalence result.

*Comparison with related work.* Earlier multimodal CXR studies typically report gains from a single run on ICU or inpatient cohorts (Hayat et al, 2022; Soenksen et al, 2022; Lee et al, 2024). We instead study ED pneumonia under a verified timing constraint, with replication across seeds and explicit audits, and reach a negative but adequately powered conclusion about discrimination. Our missingness result complements the literature on informative missingness: where GRU-D and similar models *use* missingness as a predictive feature (Che et al, 2018), we show that in a fusion model the same signal can be mistaken for a benefit from the laboratory values, and we give a simple control, keeping only the flags, that separates the two. Our calibration result agrees with benchmarks showing that the uncertainty of a single model is unreliable under distribution shift, whereas averaging several models is more robust (Ovadia et al, 2019). Comparing raw numbers across different cohorts and label definitions

is not informative; the contribution is the rigour of the evaluation and the two cautions.

*Limitations.* Training and the multimodal evaluation use a single institution (MIMIC) and are retrospective. The external validation confirms that the image model generalises to NIH, but it is limited to imaging: to our knowledge no public dataset links ED triage, or early labs, to the chest radiograph, so the fusion models themselves cannot yet be validated externally, and building such a linked multi-site resource is an important community need. The sharp fall in prevalence-dependent calibration on NIH (from 45% to 1.3%) is expected rather than a defect: before deployment in a population with a different base rate, a simple recalibration (temperature or Platt scaling) fitted on a small labelled sample from the target site would restore calibrated probabilities without retraining, and our temperature-scaling result shows that one parameter is enough internally. The subcohort with labs present is small ( $n=269$ ), but this reflects the clinical reality that laboratory results are rarely available at the moment of the first ED radiograph (Section 4.5) rather than a sampling artefact, and it is the right denominator for the question of whether labs help when they are available. Finally, the *u.ignore* policy raises the internal prevalence to 45.3%, and the NIH labels are text-mined, which adds noise to the external estimate.

## 6 Conclusion

Under a verified timing constraint and replication across seeds, combining the clinical context available at triage with the chest radiograph does not reliably improve pneumonia discrimination, and the one apparent gain comes from the pattern of missing data. The image model itself is robust and generalises across institutions with an AUROC drop of only 0.016. Beyond the negative result, the study contributes a reproducible, time-safe multimodal ED pipeline and two reusable cautions, calibration inflation from a single run and missingness as signal, for the growing field of clinical multimodal AI. Building on the ViT-Small check reported here, future work includes fine-tuning larger medical foundation models, validation across sites, and prospective evaluation with recalibration to the local base rate.

## Declarations

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*Authors' contributions.* YAD designed and implemented the data pipeline, trained all models and performed the analyses. AME conceptualised and supervised the study and contributed to the manuscript. MM and SS contributed to methodology, interpretation and critical revision. All authors read and approved the final manuscript.

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*Conflict of interest.* The authors declare no competing interests.

*Ethics approval.* All data are publicly available and de-identified (MIMIC under PhysioNet's credentialed Data Use Agreement; NIH ChestX-ray14 public release). No additional ethical approval was required; procedures followed the Declaration of Helsinki.

*Data and code availability.* MIMIC and NIH ChestX-ray14 are publicly available (MIMIC via PhysioNet credentialing). All pipeline, training and evaluation code is available at <https://github.com/wa5rx3/MultiModalPneumonia>; every reported number is regenerated by a named script tied to the fixed temporal split.

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